INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

BRAKE, ELECTRO-MECHANICAL, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high impact (H.I.) shockproof, fail-safe, electrically released, spring-set brakes for shipboard hull and deck machinery applications.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59125 - Terminal Boards, Molded, Barrier Screw and Stud Types and Associated Accessories

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-917 - Electric Power Equipment, Basic Requirements for

MIL-DTL-2212 - Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices

MIL-DTL-15109 - Resistors and Rheostats, Naval Shipboard

MIL-DTL-24093 - Steel Forgings, Carbon and Alloy Heat-Treated

MIL-S-24235 - Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, General Specification for

MIL-C-24707 - Castings, Ferrous, General Specification for

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6110

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MIL-DTL-24707/1 - Castings, Ferrous, for Machinery and Structural Applications

MIL-C-24707/5 - Castings, Ductile Iron and Austenitic Ductile Iron

MIL-DTL-32529 - Power Electronic Conversion Equipment, Naval Shipboard

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

MIL-STD-889 - Galvanic Compatibility of Electrically Conductive Materials

MIL-STD-1399-300-1 - Low Voltage Electric Power, Alternating Current

MIL-STD-1472 - Human Engineering

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume I

MIL-HDBK-781 - Reliability Test Methods, Plans, and Environments for Engineering, Development Qualification, and Production

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AQ-GIB-010/248 - Requirements for Welding and Brazing Procedure and Performance Qualification

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

T9074-BD-GIB-010/300 - Base Materials for Critical Applications: Requirements for Low Alloy Steel Plate, Forgings, Castings, Shapes, Bars, and Heads of HY-80/100/130 and HSLA-80/100

(Copies of these documents are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

ASTM INTERNATIONAL

ASTM A216/A216M - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service

ASTM A757/A757M - Standard Specification for Steel Castings, Ferritic and Martensitic, for Pressure-Containing and Other Applications, for Low-Temperature Service

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 General requirements. Brakes shall be in accordance with MIL-DTL-917. In the event of conflict, the following order of precedence shall prevail:

a. The requirements of this specification.

b. The applicable requirements of Government specifications referenced herein.

c. Non-Government standards specified herein.

3.2.1 Insulation. Insulation shall be in accordance with class B of MIL-DTL-24093, except that coils may be class A, B, or H as specified (see 6.2). Direct current (DC) brakes shall be capable of withstanding dielectric tests in accordance with MIL-STD-1399-300-1. Alternating current (AC) brakes shall be capable of withstanding dielectric tests in accordance with MIL-STD-1399-300-1. DC shunt coils not protected by a shunt resistor or other equally effective means shall be capable of withstanding, for a period of not less than 1 second, a dielectric test voltage of 10 times rated circuit voltage plus 1,000 volts. Rated voltage for submarine applications is the nominal voltage.

3.2.2 Painting. All parts not having a corrosion-resisting treatment or not fabricated of the types of corrosion-resisting materials specified in MIL-DTL-917 shall be painted in accordance with MIL-DTL-917. Touching up shall be permitted for marks or scratches due to factory handling.

3.2.3 Bearings, bushings, and pivot pins. Bearings, bushings, and pivot pins shall be of bronze, corrosion-resisting metal, oil-impregnated insert material, or other similar materials. Non-austenitic corrosion-resisting steel may be used for hardened pins. Bushings shall be self-lubricating wherever practical. Rotating anti-friction bearings shall be sealed and shall be selected to result in an L-10 life of not less than 10,000 hours.

3.2.4 Material and components. Material shall be as specified herein. Components that are in or supported by the Federal supply system shall be utilized to the fullest extent possible.

3.2.4.1 Metals. Metals used in the construction of the electro-mechanical brakes shall be, to the greatest extent possible, selected from the following:

a. Steel plate. High yield strength steel plate shall be in accordance with T9074-BD-GIB-010/300. All other steel plate shall be in accordance with MIL-DTL-24093.

b. Forged steel. Forged steel to be welded shall be in accordance with class H, type V of MIL-DTL-24093.

c. Cast steel. Cast steel to be welded shall be in accordance with MIL-C-24707 and MIL-DTL-24707/1 (grade WCA of ASTM A216/A216M or grade A1Q of ASTM A757/A757M).

d. Cast iron. Brake wheels may be of cast iron. Ductile iron in accordance with MIL-C-24707/5 may be used for housings, pressure plates, end bells, and other structural parts when approved by the Technical Authority (see 6.5.2).

3.2.4.2 Castings. Castings shall be in accordance with S9074-AR-GIB-010/278.

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3.2.4.3 Prohibited materials. The brakes shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs. The following materials are restricted in use:

a. Brittle material. Brittle material (see 6.5.1) shall not be used, except as specified for a particular application.

b. Aluminum. T-6061-T6 and 70xx-T6 series aluminum alloys shall not be used for welded load bearing fittings or components.

3.2.4.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.4.5 Dissimilar metals. Coupled dissimilar metals shall be protected against corrosion in a marine environment in accordance with MIL-STD-889.

3.3 Design.

3.3.1 Operation. Brakes shall be either the shoe or disc type as specified (see 6.2) and shall be coil operated. Brakes shall be spring-set and electrically released.

3.3.1.1 Fail-safe operation. The brake shall automatically apply the specified resisting static torque (see 6.2) to the connected load when electrical power to the brake is removed.

3.3.2 Static torque. The static torque of the brake shall be as specified (see 6.2). The brake hub shall not rotate when the brake is de-energized and the specified static torque is applied to the brake hub.

3.3.3 Speed. Unless otherwise specified (see 6.2), the brake rotating parts shall be compatible with a connected shaft speed of 1,800 revolutions per minute (rpm).

3.3.4 Electric power. AC brakes shall be designed to operate from shipboard AC power that has steady state and transient voltage and frequency characteristics in accordance with MIL-STD-1399-300-1. Those DC brakes that operate from shipboard DC power shall have steady state and transient voltage characteristics in accordance with MIL-STD-1399-300-1. Unless otherwise specified (see 6.2), the brakes shall operate at one of the following voltages:

a. AC:

(1) 110 volts, 60 Hertz, single-phase

(2) 220 volts, 60 Hertz, single-phase

(3) 440 volts, 60 Hertz, single-phase

b. DC:

(1) 115 volts

(2) 230 volts

(3) 250 volts (175 to 355)

3.3.5 Duty. The duty rating for cyclic operation of the brake shall be limited by the thermal capacity (see 6.5.3) of the brake and the maximum allowable coil cycle rate.

3.3.5.1 Thermal capacity. The minimum brake thermal capacity shall be specified (see 6.2) and derived by the following formula for the given application:

$$TC = (E_T * n)/550$$

where (E_T) is the total kinetic energy of all moving elements to be absorbed by the brake and (n) is the required number of brake cycle stops per minute (see 6.2), not less than 1. Note that (n) cannot exceed the maximum allowable coil cycle rate (see 3.3.5.2). The brake shall be capable of fully absorbing and dissipating this energy with a normal ambient temperature of not more than 72 °F (22 °C). Thermal capacity may be de-rated by 0.007 horsepower seconds per minute (hp-sec/min) per 1.8 °F (1 °C) above normal ambient. Brakes mounted in an orientation 15 degrees or more from horizontal shall be further de-rated as specified by the brake manufacturers, if required.

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3.3.5.2 Coil cycle rate. The coil cycle rate limits the number of engagements per minute of a static or holding duty brake. The required coil cycle rate for the application shall be as specified (see 6.2). If not specified, the minimum acceptable rate shall be 12 cycles per minute. The brake coil shall be rated for continuous duty.

3.3.5.3 Electrical operation. The brake shall set and apply the specified static torque (see 6.2) within 100 milliseconds of electric power being removed. The brake shall release and allow the hub to rotate freely within 100 milliseconds of electrical power being applied.

3.3.6 Manual release feature. All brakes shall be furnished with a side- or end-mounted, manually operated release mechanism as specified (see 6.2). Access to, and operation of, the brake manual release shall not require the use of tools or removal or disassembly of any part of the brake. The brake manual release shall be hand operated and shall be operable by one person without exceeding the manual force limits of MIL-STD-1472.

3.3.6.1 Manual release actuator. The brake manual release shall be hand operated through a single handle, knob, or lever. The release actuator shall remain installed on the brake without interference to brake operation or hazard to personnel.

3.3.6.2 Manual release spring return feature. The brake manual release shall be spring returned such that the brake automatically applies the specified static torque (see 6.2) to the connected load when manual input to the brake release is discontinued. Unless otherwise specified (see 6.2), the brake manual release shall not be detented or latching.

3.3.7 Enclosures. Brake housing and ancillary enclosures shall be in accordance with MIL-STD-108 and shall be one of the following types, as specified (see 6.2):

a. Drip-proof (45 degrees)

b. Watertight

c. Submersible (15 feet)

d. Explosion-proof

3.3.7.1 Through shaft feature. When specified (see 6.2), the brake shall be designed to allow the load shaft to axially extend completely though the brake assembly.

3.3.7.2 Moisture drain. Watertight and submersible enclosures shall have a moisture drain in the bottom. The opening shall be closed with a ¼-inch or larger pipe plug.

3.3.8 Mounting method. Brakes shall be motor mounted, gearbox mounted, or bedplate mounted, as specified (see 6.2). The attachment of the brake shall not unduly complicate the removal of the rotor of the motor. Applications requiring the brake to be permanently mounted in an orientation 15 degrees or more from horizontal shall be specified (see 6.2).

3.3.8.1 Shaft interface. The brake hub shall be sized and machined to fit the shaft and key dimensions specified (see 6.2).

3.3.9 Adjustment. The brake shall be self-adjusting to compensate for brake lining wear. Any use of tools to provide manual adjustments to the brake shall be those listed in the GSA Global Supply Catalog.

3.3.10 Inspection features. Unless otherwise specified (see 6.2), the brake enclosure (see 3.3.7) shall be provided with indicators, inspection ports, or other features that enable maintenance personnel to determine by inspection if the manufacturer's recommended allowable brake lining wear limit is reached, or the brake is not adjusted to within the manufacturer's recommended tolerances. The inspection features shall allow for these inspections to be accomplished without requiring operation, removal, or disassembly of any load bearing portion of the brake.

3.3.10.1 Brake lining wear indicating switch. If specified (see 6.2), a minimum thickness brake lining switch shall be provided. The switch shall be in accordance with the requirements of MIL-DTL-2212 with normally open and normally closed contacts provided. The contacts shall be rated for the control voltage in accordance with 3.3.4 and specified in 6.2.

3.3.11 Corrosion resistance. The brake shall not be vulnerable to internal corrosion and shall not require internal anti-corrosion preservatives that could interfere with brake performance.

3.3.12 Ambient temperature and humidity. Brakes shall operate as specified herein throughout an ambient temperature range of 32 to 122 °F (0 to 50 °C) and an ambient relative humidity range of 5 to 95 percent. For ambient

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temperatures above 72 °F (22 °C), the thermal capacity (see 3.3.5.1) may be de-rated. Maximum ambient temperature for application shall be as specified (see 6.2).

3.3.13 Temperature rise. Temperature rises shall not exceed the values specified in table I, as adjusted in accordance with 3.3.13.1 through 3.3.13.5.

TABLE I. Maximum temperature rise °F (°C).

Name of part	Temperature rise (122 °F [50 °C] ambient)
Shunt coils (AC):	
Class A insulation	140 (60)
Class B insulation	185 (85)
Class H insulation	302 (150)
Shunt coils (DC):	
Class A insulation	131 (55)
Class B insulation	176 (80)
Class H insulation	293 (145)
Bolted connections and terminal connections:	
Not silver plated	122 (50)
Silver plated	149 (65)

3.3.13.1 Hot-spot temperatures. Table I is based on limiting hot-spot temperatures of 230 °F (110 °C) for class A, 275 °F (135 °C) for class B, and 392 °F (200 °C) for class H insulation.

3.3.13.2 Method of measurement. Shunt coil rises shall be measured by the resistance method. All other rises shall be measured by thermocouple or thermometer.

3.3.13.3 Temperature gradient. If cotton or glass served wire or layer insulation is used, the allowable temperature rise for coils should be 9 °F (5 °C) less than that shown in table I.

3.3.13.4 Enclosure differential. Coils in watertight, submersible, or explosion-proof enclosures should be an additional 9 °F (5 °C) rise over the values specified in table I.

3.3.13.5 Submarine applications. Coils for 250-volt brakes are permitted temperature rises of 18 °F (10 °C) higher than specified in table I, as modified by 3.3.13.3 and 3.3.13.4.

3.3.13.6 Temperature sensor. If specified (see 6.2), a maximum temperature switch shall be provided and adjusted considering the operability conditions of 3.3.5.1 and table I. The switch shall be in accordance with the requirements of MIL-DTL-2212 with normally open and normally closed contacts provided. The contacts shall be rated for the control voltage in accordance with 3.3.4 and specified in 6.2.

3.3.14 Parts.

3.3.14.1 Coils. Coils for disc brakes shall be encapsulated. Closed terminal lugs with binding screws shall be used for coil wire leads. The minimum wall thickness of encapsulating materials shall be 0.031 inch. If the coil is wound on a core tube before encapsulation, the wall thickness may be reduced to 0.016 inch in the core. Coils, where of the removable type, shall be readily removable from the pole or yoke without damage to the coil. The coil shall be rigidly held so as to prevent damage to the coil leads. Coils shall be marked for ready identification.

3.3.14.2 Resistors. Resistors shall be in accordance with MIL-DTL-15109. Resistors shall be so constructed and installed that each unit can be readily replaced. Resistors shall be mounted within the brake or control enclosure. Provision shall be made for readily connecting the leads to the resistor steps and for preventing the leads from coming in contact with the heated resistor.

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3.3.14.3 Springs. Springs shall be designed to ensure the proper functioning of the equipment to which they are attached. Springs shall not be stressed beyond their fatigue limits.

3.3.14.4 Terminal leads. All terminal leads shall be made of flexible, stranded, insulated wire compatible with temperatures defined in table I and provided with closed terminal lugs. Polyvinyl chloride (PVC) insulation shall not be used on terminal leads. An insulated sleeve shall be provided to prevent abrasion of the lead insulation by metallic edges of the frame, terminal box, or cable clamp; other methods to prevent abrasion shall be approved by the appropriate Technical Authority (see 6.5.2).

3.3.14.5 Terminal boards. The brake shall include a terminal board for connection of the coil leads to all external wiring connections. Terminal boards shall be in accordance with A-A-59125.

3.3.14.6 External cable feed through. Feed through for connections for external cabling shall be in accordance with MIL-S-24235.

3.3.14.7 Wire marking. All wires shall be permanently identified in accordance with MIL-DTL-917 at each termination point by wire markers of synthetic resin tubing or fiber tags. Markers shall identify cable number, wire number, and connection point.

3.3.14.8 Threaded fasteners. Tools used on fasteners shall be those listed in the GSA Global Supply Catalog. Fastener material selection and interfacing bolted surfaces shall be compatible in accordance with 3.2.4.5. External threaded fasteners shall be provided with a corrosion control method that does not rely on a painted coating.

3.3.15 Operational requirements.

3.3.15.1 Inclination. The brake shall operate without damage or degradation when inclined in any direction at an angle of 45 degrees from the normal mounting position (see 3.3.8).

3.3.15.2 Vibration. The brake shall operate without damage or degradation when subjected to the vibration requirements of MIL-STD-167-1, type I and type II.

3.3.15.3 Shock. Brakes shall meet the requirements of MIL-DTL-901 for grade A or grade B, as specified (see 6.2), in the engaged position for equipment in the stowed configuration. MIL-DTL-901 grade A brakes shall operate satisfactorily before and after the periods of shock. Compliance with the MIL-DTL-901 requirement shall not depend upon shock isolation mounts or the use of a device or devices by which the operator can lock the brake in position.

3.3.15.4 Electromagnetic compatibility. The brake shall meet and demonstrate compliance with MIL-STD-461.

3.4 Identification plates. Each brake shall be provided with an identification plate mounted on an exterior part not renewed during its life. Plates shall be brass, corrosion-resistant steel, or anodized aluminum of not less than 0.030 inch in thickness and shall be marked by etching, engraving, stamping, or photo-etch process. Etching, engraving, or stamping shall be not less than 0.003 inch deep. The identification plates for brakes shall be marked to include, as a minimum, the following:

a. Manufacturer’s name and CAGE code
b. Manufacturer’s model, part number, and serial number
c. Manufacturer drawing number and revision
d. Coil voltage, frequency, current rating and duty
e. Static torque
f. Enclosure type
g. Class of insulation
h. Thermal capacity
i. Assembled weight
j. Rotational inertia

3.5 Workmanship. Neatness, thoroughness, and inspection of welding shall be in accordance with S9074-AR-GIB-010/278. Particular attention shall be given to neatness and thoroughness of marking of parts and

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subassemblies, support for electric cables, squareness and parallelism of butting joints, and absence of burrs and sharp corners on all parts.

3.6 Human engineering. The brake shall be designed for installation, operation, and maintenance by individuals from the 5th percentile female through 95th percentile male population using the anthropometric and strength requirements of MIL-STD-1472.

3.7 Reliability. The brake shall not require replacement of component parts, except brake linings, during the mean cycles between failures specified when operated at the duty rate (see 3.3.5).

3.8 Maintainability. Unless otherwise specified (see 6.2), the maximum mean time to repair shall be 4 hours. Verification of proper brake adjustment, inspection for brake wear, and cleaning/removal of dust resulting from brake operation shall require the use of tools listed in the GSA Global Supply Catalog.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. One brake shall be subjected to first article inspection when a first article sample is required (see 6.2). This inspection shall include the examinations and tests in table II.

4.3 Conformance inspection. Each brake shall be subjected to conformance inspection. This inspection shall include the examinations and tests in table II.

TABLE II. First article and conformance inspection.

Characteristic	Requirement	Verification	First article	Conformance
Insulation	3.2.1	4.4.1	X	X
Painting	3.2.2	4.4.2	X	X
Bearings	3.2.3	4.4.3	X	X
Material	3.2.4	4.4.4	X	X
Prohibited materials	3.2.4.3	4.4.4.1	X	X
Operation	3.3.1	4.4.5	X	X
Fail-safe	3.3.1.1	4.4.5	X	X
Static torque	3.3.2	4.4.6	X	X
Speed	3.3.3	4.4.7	X	--
Electric power	3.3.4	4.4.8	X	--
Thermal capacity	3.3.5.1	4.4.9	X	--
Coil cycle rate	3.3.5.2	4.4.10	X	--
Electric operation	3.3.5.3	4.4.11	X	X
Manual operation	3.3.6	4.4.12	X	X
Enclosures	3.3.7	4.4.13	X	--
Through shaft	3.3.7.1	4.4.13	X	X
Moisture drain	3.3.7.2	4.4.13	X	X
Mounting	3.3.8	4.4.14	X	--
Shaft interface	3.3.8.1	4.4.14	X	X
Adjustment	3.3.9	4.4.15	X	X

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TABLE II. First article and conformance inspection – Continued.

Characteristic	Requirement	Verification	First article	Conformance
Inspection features	3.3.10	4.4.16	X	X
Brake switch	3.3.10.1	4.4.17	X	X
Corrosion resistance	3.3.11	4.4.18	X	X
Ambient temperature/humidity	3.3.12	4.4.19	X	--
Temperature rise	3.3.13	4.4.20	X	--
Temperature sensor	3.3.13.6	4.4.21	X	X
Coils	3.3.14.1	4.4.22.1	X	X
Resistors	3.3.14.2	4.4.22.2	X	X
Springs	3.3.14.3	4.4.22.3	X	X
Leads	3.3.14.4	4.4.22.4	X	X
Terminal boards	3.3.14.5	4.4.22.5	X	X
Feed through	3.3.14.6	4.4.22.6	X	X
Wire marking	3.3.14.7	4.4.22.7	X	X
Threaded fasteners	3.3.14.8	4.4.22.8	X	X
Inclination	3.3.15.1	4.4.23	X	--
Vibration	3.3.15.2	4.4.24	X	--
Shock	3.3.15.3	4.4.25	X	--
Electromagnetic compatibility	3.3.15.4	4.4.26	X	--
Identification plates	3.4	4.4.27	X	X
Workmanship	3.5	4.4.28	X	X
Human engineering	3.6	4.4.29	X	--
Reliability test	3.7	4.4.30	X	--
Maintainability demonstration	3.8	4.4.31	X	--

4.4 Verification methods.

4.4.1 Insulation. It shall be verified by inspection that appropriate class insulation for application ambient temperature is utilized (see 6.2). Coil insulation resistance shall be measured and corrected to 77 °F (25 °C). Insulation resistance shall be not less than 10 megohms.

4.4.2 Painting. It shall be verified by inspection that all parts not having a corrosion-resisting treatment or not fabricated of corrosion-resisting materials shall have been painted in accordance with MIL-DTL-917.

4.4.3 Bearings. It shall be verified by inspection that all bearings, bushings, and pivot pins are fabricated of corrosion-resistant materials. It shall be verified that bearings are sealed and that, where possible, bushings are also sealed.

4.4.4 Material. Materials of fabrication shall be visually inspected for compliance with requirements of 3.2.4.

4.4.4.1 Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.4.3 and 6.2).

4.4.5 Operation. It shall be demonstrated that, when the brake release coil(s) are electrically energized, the brake is fully released. It shall be demonstrated that when the electrical source is de-energized, the brake is spring-set to engage (fail-safe).

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4.4.6 Static torque test. Brakes shall be subjected to a static torque test as follows to verify that the static holding torque delivered by the brake equals or exceeds that specified (see 6.2):

a. Apply a torque equal to the specified static torque (see 6.2) to the brake. Verify that the brake hub does not rotate when the specified static torque (see 6.2) is applied.

b. Repeat step 4.4.6.a, but with torque applied in the opposite direction.

c. Manually release the brake, rotate the brake hub 90 degrees, and then reset the brake. Verify that the brake hub rotates freely while released.

d. Repeat steps 4.4.6.a and 4.4.6.c three more times to verify static holding torque at each hub rotational position to demonstrate performance and repeatability in each direction of operation.

4.4.7 Speed. Brake release coils shall be energized and operation of the brake shall be demonstrated at maximum speed as specified (see 6.2) for 10 minutes in each direction. The brake shall not display evidence of unusual noise or heat.

4.4.8 Electric power. Those DC brakes that are provided power from shipboard DC power supply shall be tested in accordance with MIL-DTL-32529. Failure of a DC brake to release at 80 percent of rated voltage for DC when cold (room ambient) shall be cause for rejection. AC brakes shall be tested in accordance with MIL-STD-1399-300-1. Failure of an AC brake to release at 85 percent of rated voltage for AC at rated frequency when cold (room ambient) shall be cause for rejection.

4.4.9 Thermal capacity. It shall be verified by calculation that the thermal capacity rating of the brake meets the maximum kinetic stopping requirement and the maximum number of brake cycle stops required per minute (see 6.2).

4.4.10 Coil cyclic rate. It shall be verified by inspection that the coil cyclic rating exceeds the application requirement for number of brake cycle stops per minute (see 6.2).

4.4.11 Electric operation (static torque). The operation of the brake electric release shall be demonstrated as follows:

a. Apply a torque equal to the specified static torque (see 6.2) to the brake. Verify that the brake hub does not rotate when the specified static torque (see 6.2) is applied.

b. Apply electric power of the specified voltage and frequency to the brake coil. Verify that the coil energizes and the brake releases. Verify that the brake release time does not exceed that specified in 3.3.5.3. Rotate the hub ¼ turn.

c. Remove electric power from the brake coil. Verify that the brake resets and the brake hub does not rotate when the specified static torque (see 6.2) is applied. Verify the brake reset time does not exceed that specified in 3.3.5.3.

d. Repeat steps 4.4.11.a through 4.4.11.c nine more times, reversing the direction of rotation each time to demonstrate performance and repeatability in each direction of operation.

4.4.12 Manual release operation. The operation of the brake manual release shall be demonstrated as follows:

a. Apply a torque equal to the specified static torque (see 6.2) to the brake.

b. Operate the manual brake release. Verify that, when operated, the brake manual release allows the input shaft to rotate. Verify that the manual input required to operate the brake release does not exceed the force limits specified by MIL-STD-1472. Rotate the hub ¼ turn.

c. Discontinue manual input to the brake manual release. Verify that the brake manual release returns to its normal position, that the brake resets, and that the brake hub does not rotate when the static torque load is applied.

d. Repeat steps 4.4.12.a through 4.4.12.c seven more times, reversing the direction of rotation each time to demonstrate performance and repeatability in each direction of operation.

4.4.13 Enclosures. Enclosures shall be tested and inspected for compliance with the requirements of 3.3.7, as applicable, for the specified degree of tightness (see 6.2). It shall be verified by inspection that, if specified (see 6.2), the through hole provides clearance for the shaft diameter. It shall be verified that the drain plug, if required (see 3.3.7.2), is installed.

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4.4.14 Mounting. It shall be verified by inspection that the brake assembly is configured to mount as specified (see 6.2). For brakes that are specified to be mounted at an angle greater than 15 degrees from the horizontal (see 6.2), confirm by inspection that the design precludes the possibility of dragging friction material when released. It shall be verified by inspection that the brake bore is machined to interface with the shaft diameter as specified (see 6.2).

4.4.15 Adjustment. It shall be verified by inspection that the brake design is provisioned with a self-adjusting mechanism to compensate for brake friction material wear. Also, it shall be verified that all adjustment points may be performed with the use of tools listed in the GSA Global Supply Catalog.

4.4.16 Inspection features. It shall be verified by inspection that the design allows for in-service maintenance inspection of equipment that would require adjustment.

4.4.17 Brake lining wear switch. It shall be verified by demonstration that the switch actuates when the brake lining thickness is below minimum allowable thickness.

4.4.18 Corrosion resistance. It shall be verified by inspection that brake interior design and material selection is consistent with corrosion control without the use of materials or coating that could adversely affect brake performance.

4.4.19 Ambient temperature and humidity. It shall be verified by test in accordance with MIL-STD-810, method 507.4, that the brake design is compatible with the ambient temperature and humidity ranges specified in 3.3.12. Also, it shall be verified by calculation that if the expected ambient temperature exceeds 72 °F (22 °C), the brake can operate as specified (see 6.2) without exceeding the thermal capability (see 3.3.5.1), or if necessary, the brake shall be de-rated to suit the ambient conditions with Technical Authority approval (see 6.5.2).

4.4.20 Temperature rise. The temperature rise tests shall be as specified in 4.4.20.1 through 4.4.20.4.

4.4.20.1 General requirement. Temperature rise tests shall be made at rated voltage and frequency with the brake subjected to the duty rating specified in 3.3.5 to determine that the rises specified in table I are not exceeded. Temperature rise tests shall be made on individual parts of the completely assembled equipment. The specified voltage shall be applied to all parts simultaneously so that the mutual heating effect of the several parts are present.

4.4.20.2 Starting temperature. Temperature rise tests shall not be started if the ambient differs more than 9 °F (5 °C) from the ambient where the brake has stood during the preceding 2 hours.

4.4.20.3 Ambient variations. A variation of more than 18 °F (10 °C) during a period of 6 hours, or a proportional change for runs of shorter duration, shall in no case be permitted. If the ambient variations are excessive or if the ambient changes rapidly at the end, the test shall be repeated.

4.4.20.4 Duration of tests. The temperature rise test for continuously rated brakes shall be continued until the temperature that can be observed during the test has attained a steady final value. The temperature rise test for intermittently rated brakes shall be continued for the duty cycle specified for the brake.

4.4.21 Temperature sensor. It shall be verified by demonstration that the switch actuates when the maximum allowable operating temperature is exceeded.

4.4.22 Parts. It shall be verified by inspection that the following parts are designed and provided.

4.4.22.1 Coils. It shall be verified that the coils on disc brakes are encapsulated. It shall be verified that removable coils can be removed without damage. It shall be verified that the coils are rigidly held in the coil and that the cores are marked for ready replacement identification.

4.4.22.2 Resistors. It shall be verified that the resistors are in accordance with MIL-DTL-15109, that they are mounted inside the brake housing or control enclosure, and that they are readily replaceable. It shall be verified that the leads are dressed clear of the resistor.

4.4.22.3 Springs. It shall be verified that the springs are not closed, coiled, or loose.

4.4.22.4 Terminal leads. It shall be verified that the leads are flexible, stranded, insulated wires provided with closed terminal lugs. It shall be verified by inspection that PVC insulation is not used. Protective insulated sleeves shall be provided as necessary to protect the leads from sharp edges.

4.4.22.5 Terminal boards. It shall be verified that a terminal board, in accordance with 3.3.14.5, is provided with coil leads terminated on the board.

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4.4.22.6 External cable feed through. It shall be verified that a feed through fitting is provided in accordance with MIL-S-24235.

4.4.22.7 Wire marking. It shall be verified that all internal wiring is permanently identified in accordance with MIL-DTL-917.

4.4.22.8 Threaded fasteners. It shall be verified that all fasteners can be engaged with common shipboard tools in accordance with the GSA Global Supply Catalog.

4.4.23 Inclination. The brake shall be tested for proper operation in accordance with 4.4.5 at rated voltage and frequency when inclined at 45 degrees in the forward, backward, left, and right positions.

4.4.24 Vibration. The brake shall operate without damage or degradation when subjected to the vibration requirements of MIL-STD-167-1. During the exploratory vibration tests and variable frequency tests, the brake shall be tested in the energized (released) position for not more than 10 continuous minutes or an increase of brake temperature of more than 18 °F (10 °C) above ambient, whichever is longer, while rotating at maximum shaft speed and subjected to the input frequency requirements of type I testing. The input shaft shall then be stopped and the brake set until the brake has cooled to within 9 °F (5 °C) of ambient. Type I testing shall continue until completion of exploratory vibration tests and variable frequency tests. Endurance testing at selected prominence frequencies shall be conducted with the brake released for 50 percent of the test and engaged for the other 50 percent. Brake temperature shall be limited to an increase of 18 °F (10 °C) above ambient and allowed to cool to within 9 °F (5 °C) of ambient. Type II testing shall be conducted continuously with the brake released. If the brake exhibits a continuous thermal increase during type II testing, the test shall be stopped and the brake inspected for defects.

4.4.25 Shock. Shock tests shall be conducted in accordance with MIL-DTL-901 for grade A or grade B, as specified (see 6.2).

4.4.25.1 Rejection criteria.

a. Grade A – Breakage or visible distortion of any parts. Inability to pass the tests specified in 4.4.7, 4.4.11, and 4.4.12 after being subjected to shock.

b. Grade B – Parts of the brake coming adrift, endangering personnel or other equipment.

c. For grades A and B, any condition that results in an electrical hazard.

d. Failure to pass inspection. The brake shall be disassembled following checks (a) and (b) and inspected thoroughly for damage. The extent of disassembly need be only to the point where the condition of the brake can be easily observed. Any observed effects of the shock shall be recorded.

4.4.25.2 Disposal of shock-tested brakes. Shock-tested brakes shall be disposed of as follows:

a. Brakes that have been subjected to and successfully passed the shock test shall be considered acceptable for shipboard use, provided post-shock tests are passed and corrective measures are taken, and the shock tested article is reconditioned. Deformations that do not cause rejection but are in excess of dimensional tolerances specified on the brake drawing, shall be corrected.

b. Grade A brakes that have been subjected to the shock test and have failed structurally shall not be considered acceptable. Grade A brakes that have failed due to fastener, fittings, or other nonstructural parts may be reconditioned and resubmitted for shock tests.

c. Grade B brakes that have been subjected to the shock test and have failed in such a manner that parts have come adrift shall not be considered acceptable.

4.4.26 Electromagnetic compatibility. The brake shall be tested in accordance with MIL-STD-461. When energized, the brake shall comply with the emissions requirements of MIL-STD-461. When subjected to the susceptibility tests of MIL-STD-461, the brake shall not be damaged, rendered inoperable, or exhibit unsafe operation.

4.4.27 Identification plates. It shall be verified by inspection that identification plates are marked as specified in 3.4.

4.4.28 Workmanship. It shall be verified by inspection that any welds are compliant with S9074-AR-GIB-010/278. Also, it shall be verified that no sharp edges or burrs are present internally or externally, that all fasteners are tight, and all wiring is labeled, dressed, and anchored.

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4.4.29 Human engineering. It shall be verified by inspection that the brake design complies with the guidance of MIL-STD-1472.

4.4.30 Reliability. The reliability of the brake static hold and electric release operation shall be demonstrated by the successful completion of a reliability test using the guidance provided in MIL-HDBK-781. Applicable test level and test plan of MIL-HDBK-781 should be as specified (see 6.2). The brake shall be cycled without interruption by failure for 250 cycles at a rate of 1 cycle every 3 minutes. Each cycle shall consist of the following:

a. Apply the specified electrical power to the brake. Verify that the brake releases and allows the hub to rotate freely at 1800 rpm (±5 percent). Maintain the specified electrical power to the brake for 2 minutes.

b. De-energize the brake. Verify that the brake sets and the brake hub does not rotate.

c. After 1 minute, repeat these steps until 250 cycles have been reached.

d. After that 250th cycle, verify that the brake sets and that the brake hub does not rotate when the specified static torque (see 6.2) is applied.

4.4.31 Maintainability. The mean time to repair shall be demonstrated by the successful completion of a maintainability demonstration using the guidance provided in MIL-HDBK-470. As a minimum, the demonstration shall include the following:

a. Inspection and replacement of brake linings.

b. Inspection of brake discs or wheels for wear.

c. Inspection, adjustment, and replacement of coils and springs.

d. Internal cleaning to remove dust generated due to brake lining wear.

e. Utilization of the inspection features specified in 3.3.10.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The brake is a spring-set, electric coil-released, fail-safe brake intended for shipboard hull and deck machinery applications. These electro-mechanical brakes have design and testing requirements to meet shipboard power interface requirements and corrosion prevention. Current stock of electro-mechanical brakes procured to previous revisions of this specification may be used until depleted; future requisitions for electro-mechanical brakes should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1, 4.2, and 6.3).

c. Coil insulation (see 3.2.1 and 4.4.1).

d. Requirement for prohibited materials verification (see 3.2.4.3 and 4.4.4.1).

e. Type of brake (see 3.3.1).

f. Static torque (see 3.3.1.1, 3.3.2, 3.3.5.3, 3.3.6.2, 4.4.6, 4.4.11, 4.4.12.a, and 4.4.30.d).

g. Speed, if other than 1800 rpm (see 3.3.3 and 4.4.7).

h. Supply voltage (see 3.3.4).

i. Thermal capacity (see 3.3.5.1).

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j. Number of brake cycle stops per minute (see 3.3.5.1, 4.4.9, and 4.4.10).
k. Coil cycle rate (see 3.3.5.2).
l. Manual release location (see 3.3.6).
m. Detent specification (see 3.3.6.2).
n. Enclosure type (see 3.3.7 and 4.4.13).
o. Through shaft (see 3.3.7.1 and 4.4.13).
p. Type of mounting (see 3.3.8 and 4.4.14).
q. Mounting orientation above horizontal (see 3.3.8 and 4.4.14).
r. Shaft data (see 3.3.8.1 and 4.4.14).
r. Inspection features (see 3.3.10).
s. Whether a brake lining wear switch is to be provided (see 3.3.10.1).
t. Whether brake lining wear switch contacts are to be rated for the control voltage (see 3.3.10.1).
u. Maximum ambient temperature (see 3.3.12 and 4.4.19).
v. Whether a temperature switch is to be provided (see 3.3.13.6).
w. Control voltage to be applied to the temperature switch contacts (see 3.3.13.6).
x. Shock grade (see 3.3.15.3 and 4.4.25).
y. Mean time between failures, if other than 4 hours (see 3.8).
z. Reliability test (see 4.4.30).
aa. Packaging requirements (see 5.1).
bb. Engineering drawing technical content (see A.1.1).

6.3 First article. When a first article inspection is required (see 6.2), the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Subject term (key word) listing.

Coil
Disc
Solenoid

6.5 Definitions.

6.5.1 Brittle material. Brittle material is material showing less than 10 percent elongation in 2 inches for the standard tensile test. For the static loading case (normal ship operation), brittle material is material being used below its nil ductility transition (NDT) temperature as measured by the Naval Research Laboratory (NRL) drop weight test. Charpy V-notch impact strength values may be used as criteria when these have been correlated with drop weight test results.

6.5.2 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.5.3 Thermal capacity. The ability of a brake to absorb and dissipate heat generated solely within the brake without exceeding temperature limitations. Ambient temperature influences the ability of the brake to dissipate absorbed energy.

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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ENGINEERING DRAWING TECHNICAL CONTENT

A.1 SCOPE

A.1.1 Scope. This appendix covers information that should be included on the drawings when specified (see 6.2). This appendix is not a mandatory part of this specification. Information contained herein is for guidance only.

A.2 DRAWING CONTENTS

A.2.1 Drawing contents. The drawing should contain the following technical information features:

a. A list of materials, listing all components. Each item should be identified by the part or catalog number assigned by the original manufacturer of that item. Quantity and unit weight of each component should be identified.

b. Assembly and details with front and side view of the brake together with overall dimensions. The center of gravity should be shown in both views of the assembly.

c. Weight of the assembled brake and rotation inertia of brake rotating parts.

d. Type of mounting.

e. Torque rating of the brake.

f. Thermal capacity and coil cyclic rating of the brake.

g. Brake voltage.

h. Ambient temperature rating.

i. Manual release detail.

j. Type of enclosure.

k. Class of insulation.

l. Electric wiring and schematic diagrams.

m. Description of operation.

n. Description of adjustments which can be made on the equipment, including the manufacturer's recommended adjustment values/settings.

o. Static torque verification procedure.

p. Shock qualification letter specifying testing activity and date of approval.

q. Vibration qualification letter specifying testing activity and date of approval.

r. EMI qualification letter specifying testing activity and date of approval.

s. Drawings should be numbered to fit into logical groupings with subassemblies following assemblies, and so forth, starting with the assembly drawing.

t. Drawings should be done on computer aided drafting machines, in a language compatible with the Initial Graphics Exchange Specification (IGES) prepared by the U.S. Bureau of Standards.

u. A table of repair parts and special tools.

v. A table of labels, warnings, cautions, and instruction plates.

w. Specific installation requirements and instructions for the brake being supplied. This should include mounting surface, shaft, and key interface requirements and tolerances, lubrication requirements, and required installation, removal, and maintenance envelopes.

x. Where materials of identical or equal characteristics can be identified by more than one specification or standard, the drawings should reference only one specification or standard in accordance with the following order of preference:

(1) Industry and technical society specification or standard.

(2) Federal specification.

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(3) Military specification.

(4) Manufacturer's specification or standard.

CONCLUDING MATERIAL

Preparing Activity:
Navy – SH
(Project 6110-2025-008)

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